

Role of open source frameworks

There are many open source frameworks that play a pivotal role in implementing this solution. The Spring Boot 3.x framework provides built-in support for WebSockets and

STOMP, reducing boilerplate code. Spring WebSocket simplifies WebSocket handling with STOMP support. The Spring messaging library integrates messaging capabilities using a broker. For middleware communication, RabbitMQ

or ActiveMQ act as robust message brokers to distribute messages in high-load applications.

WebSockets enable a persistent, full-duplex connection between the client and server, allowing real-time communication. STOMP enhances WebSockets by providing message format structuring, message routing, and subscription mechanisms.

Use case: Real-time stock market updates

Consider building a stock market platform where stock prices change every second. The server broadcasts these changes in real time using STOMP over WebSockets. Clients subscribed to a particular stock receive immediate updates without refreshing the page.

First, create the application that includes the necessary dependencies in *pom.xml* (application built in Maven). This will be similar in an application built in Gradle also.

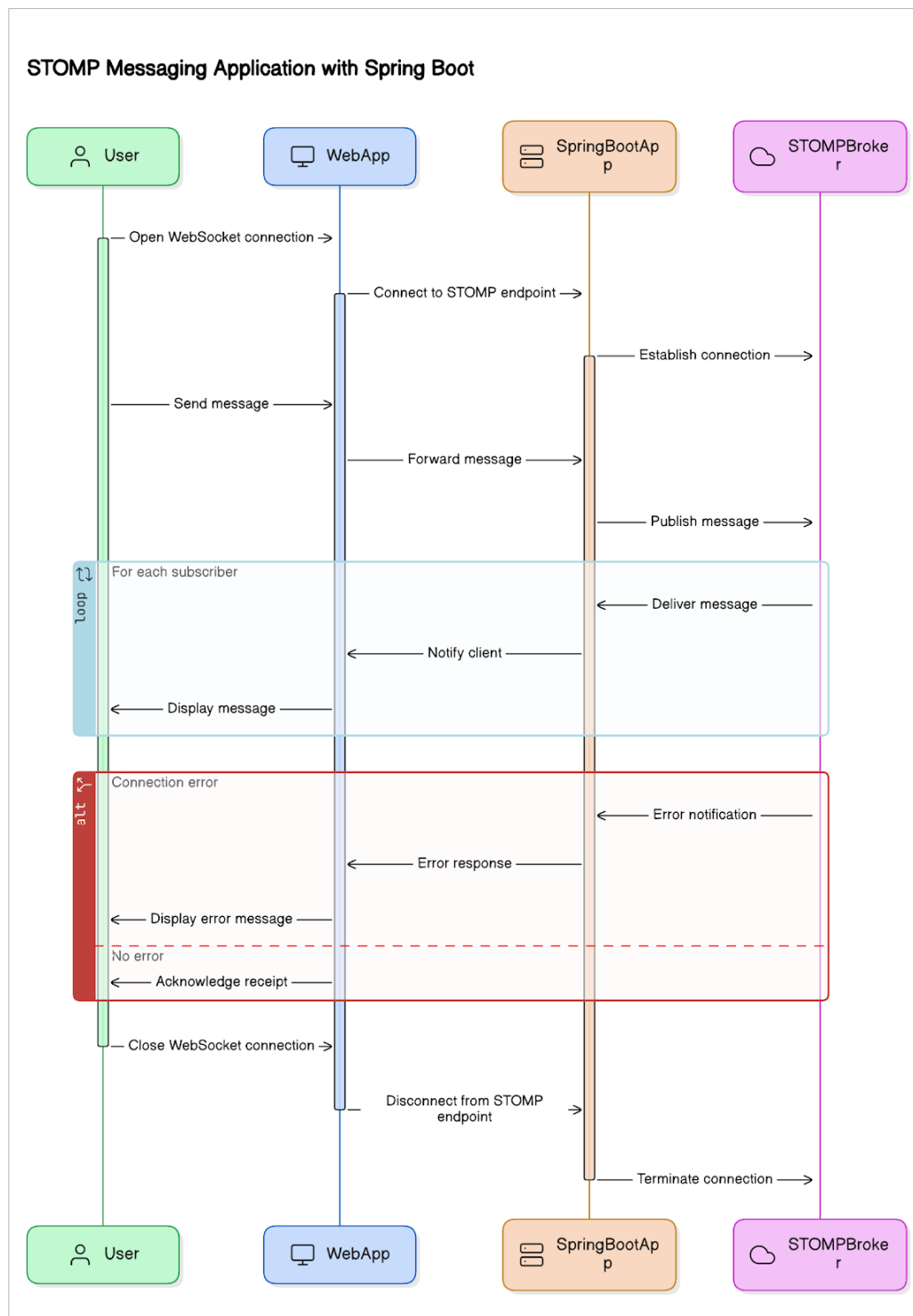


Figure 2: Sequence diagram for the STOMP based communication